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Autonomous Systems

Decision Making for Mobile Robot Navigation using AlphaBot2

Group 18
Presentation 1

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Introduction and Motivation

What is Decision Making in Robotics
Conceptualization

Process of selecting the best action for a robot to achieve a goal. Operates in sequential decision problems (not single-step decisions) and must handle uncertainty in perception and motion

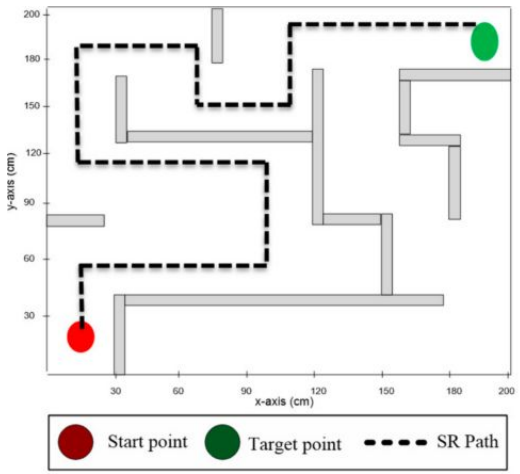
Objective → Maximize cumulative long-term reward = Minimize cumulative long-term cost

Core Challenges

- → Robot motion is uncertain (motor noise, slippage)
- → Sensor information is imperfect (limited field of view, noise)
- → If no real odometry, position must be estimated indirectly

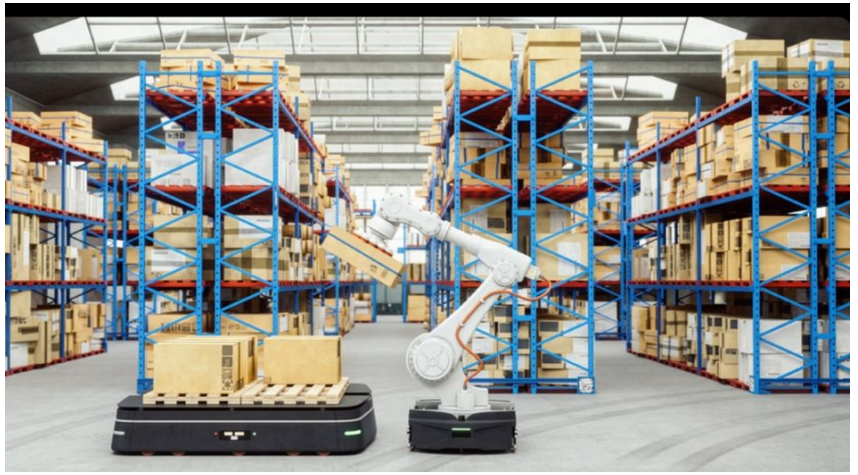
Real World Applications

Navigation in Mazes



- Finding a path from A to B in an unknown or partially known environment.
- The state space is relatively small (grid positions), the reward structure is simple.

Warehouse Automation



- Continuously executes tasks, and adapts to a changing environment.
- The state space is larger. Includes location, cargo status, shelf inventory, and time constraints.

Markov Decision Processes

A **Markov Decision Process (MDP)** is a formal mathematical model for decision making under uncertainty, where outcomes are partly random and partly controlled by the agent.

It is defined by:

- S - Set of possible **states** the robot can be in;
- A - Set of **actions** available to the robot;
- T(s, a, s') - **Transition probability** of reaching s' from s after action a;
- R(s, a) - **Reward** received for taking action a in state s;
- $\gamma \in [0,1]$ - **Discount factor**, balancing immediate vs future rewards.

The future depends only on the current state, not the history:

$$p(s_{t+1} \mid s_t, a_t)$$

A **policy** π maps states to actions telling the robot what to do in every situation.

To find the optimal policy π^*, we first compute the value function V*(s) (a measure of how good it is to be in state s assuming optimal behavior from that point onward) which is defined recursively by the **Bellman Equation**:

$$V^*(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} T(s,a,s') V^*(s') \right]$$

There are 2 classical approaches to solve it:

- **Value Iteration**: repeatedly update the value of every state until convergence;
- **Policy Iteration**: start with any policy, evaluate it, improve it and repeat until optimal.

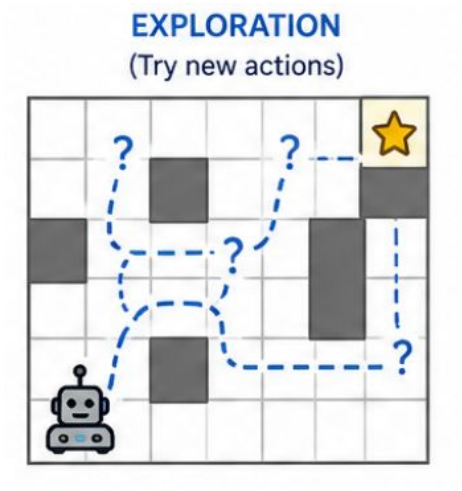
Reinforcement Learning

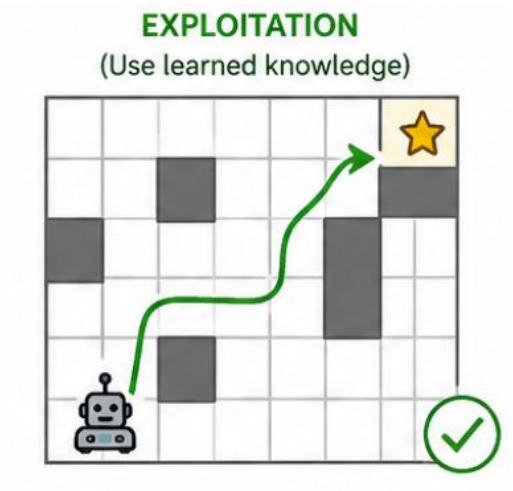
An agent learns to make decisions by interacting with an environment since is **has not prior knowledge** of it.

Unlike in MDPs, the policy must be **learned from experience** rather than computed from a **known model**.

Reinforcement Learning →

- When **transition model** T(s, a s') is **unknown**
- Robot learns optimal behavior through **trial and error**
- **No prior knowledge** of environment dynamics required





Try new actions to discover better strategies

Use current best-known actions to maximize reward

Hardware

AlphaBot2

Main Components

- Raspberry Pi 3 B
- Omni-direction wheel
- ST188: Reflective infrared photoelectric sensor
- 2 chargeable batteries (5V port)
- 5M pixels Camera
- 2x N20 micro gear motor

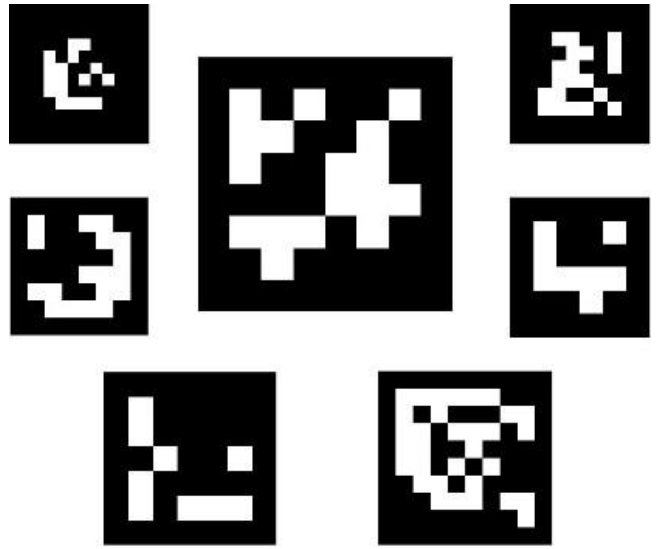
The environment is modeled as a grid where each cell represents a possible robot state (position, orientation, velocity and acceleration). The goal cell provides positive rewards, while obstacles carry negative penalties.

State Estimation Pipeline

Virtual odometry tracks motion between steps but **accumulates drift over time**

Vision System provides **position corrections** with the information from the **Visual Markers**

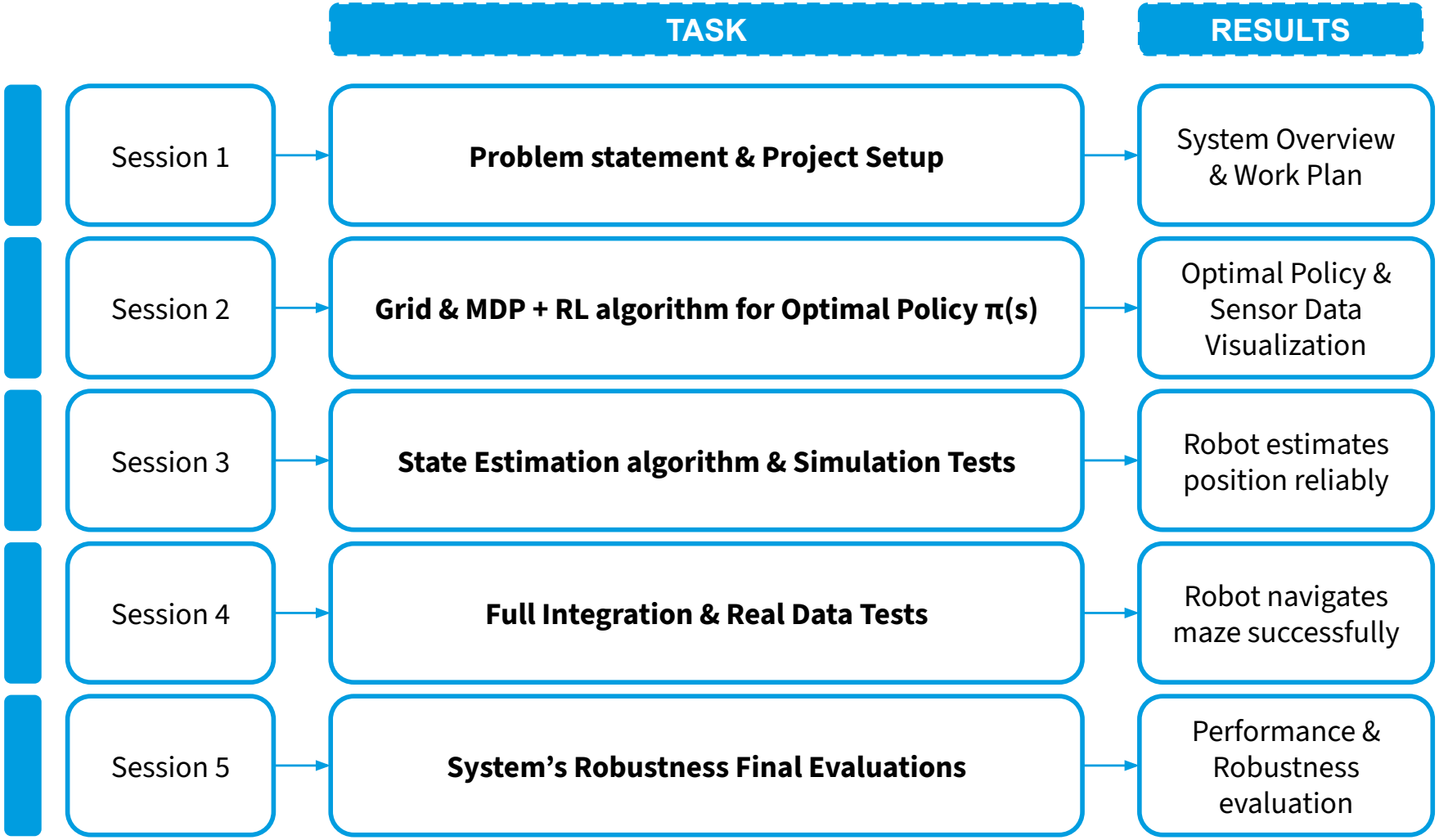
Sensor Fusion combines both sources to maintain accurate state estimates **for policy execution**



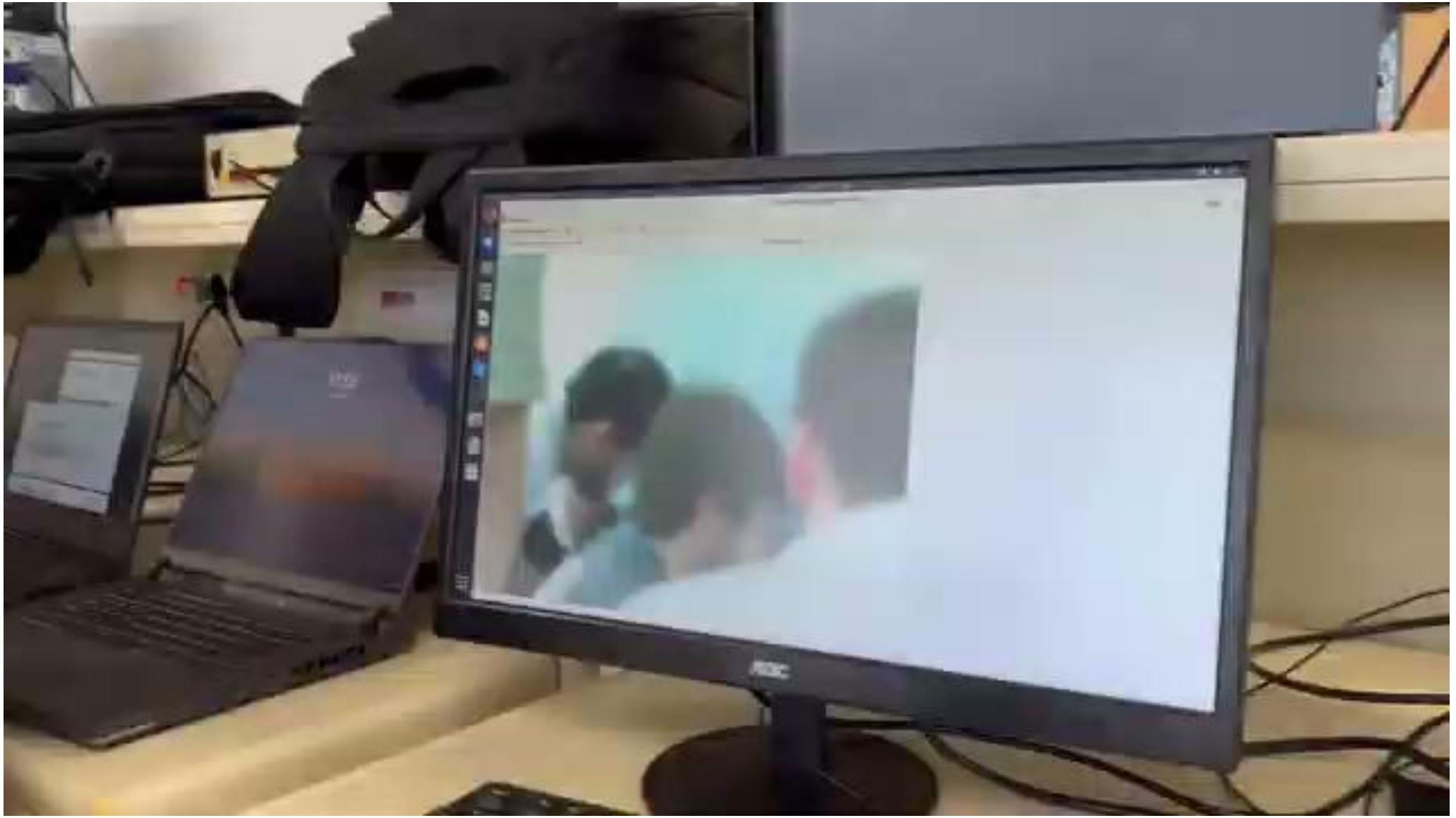
Examples of ArUco Markers

Work Plan

5-Session Work Plan



Robot Setup and Early Testing



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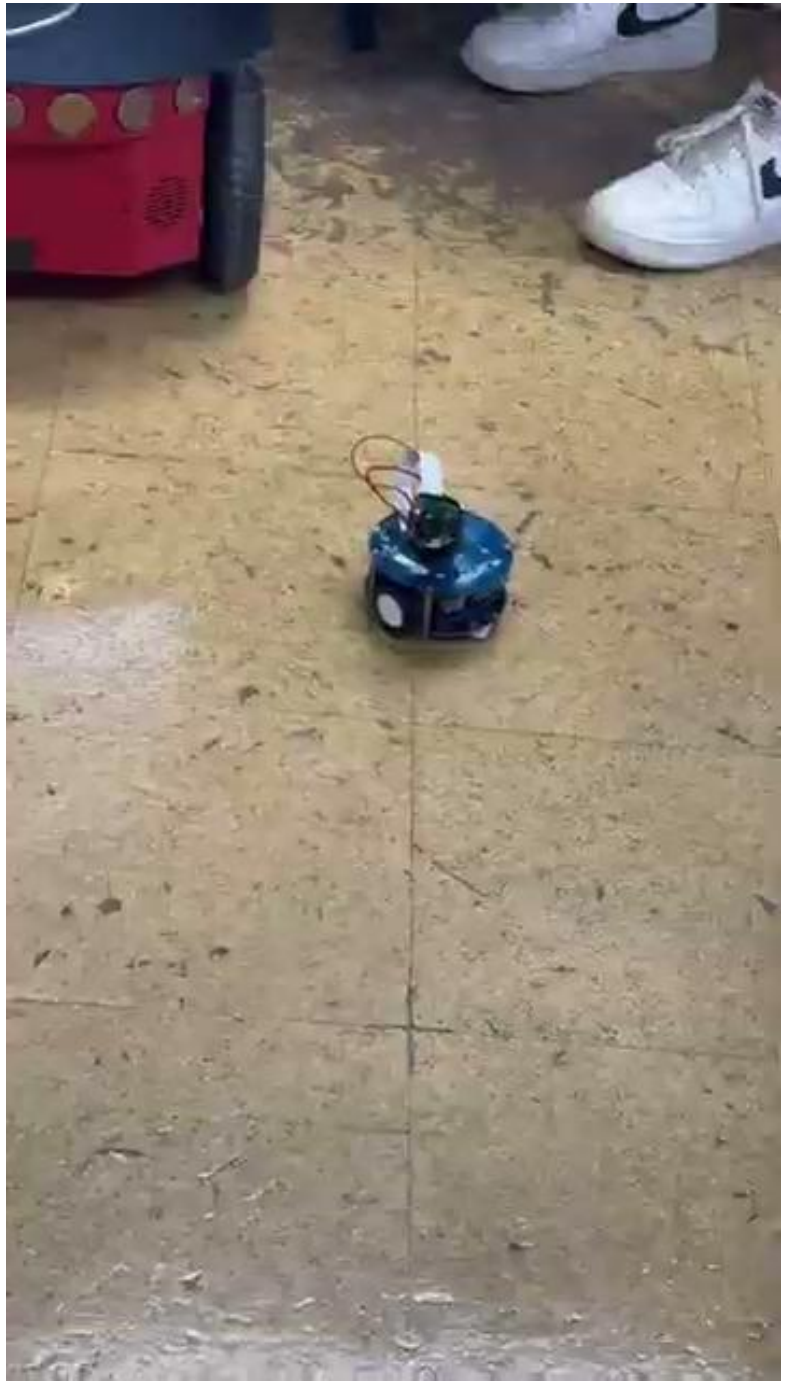

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Thank You!